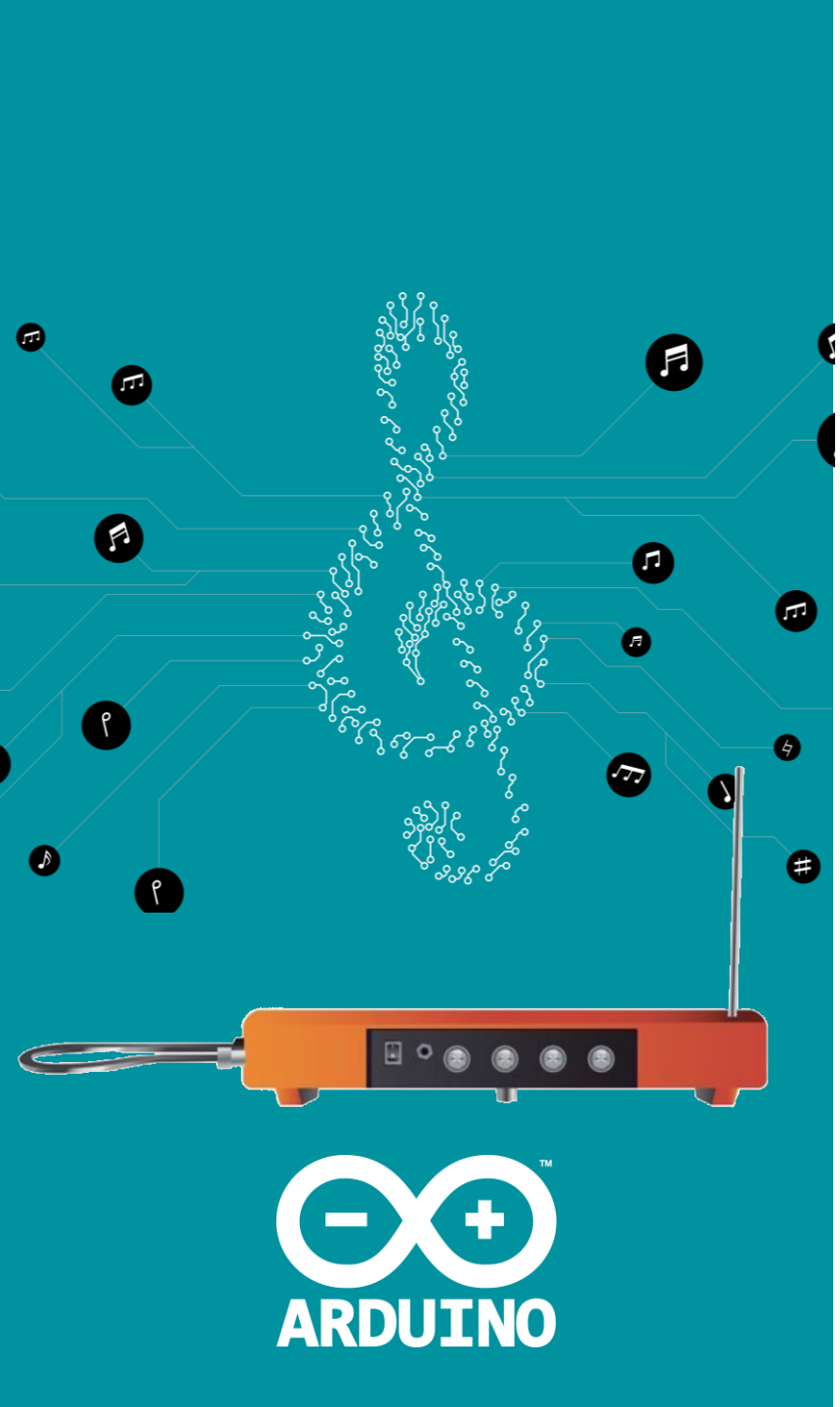


Project 6:

Light Theremin

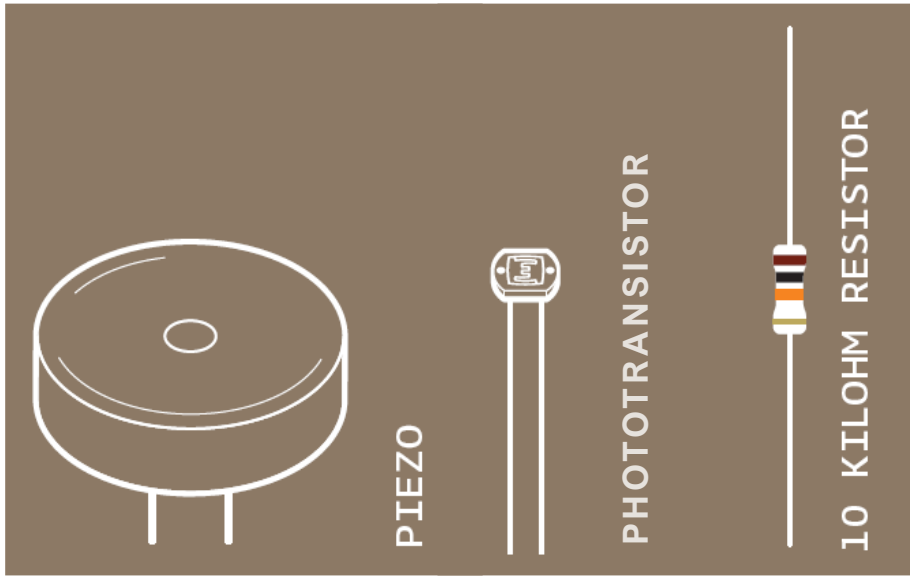
Discover: *Making sound with the tone() function, calibrating analog sensors.*

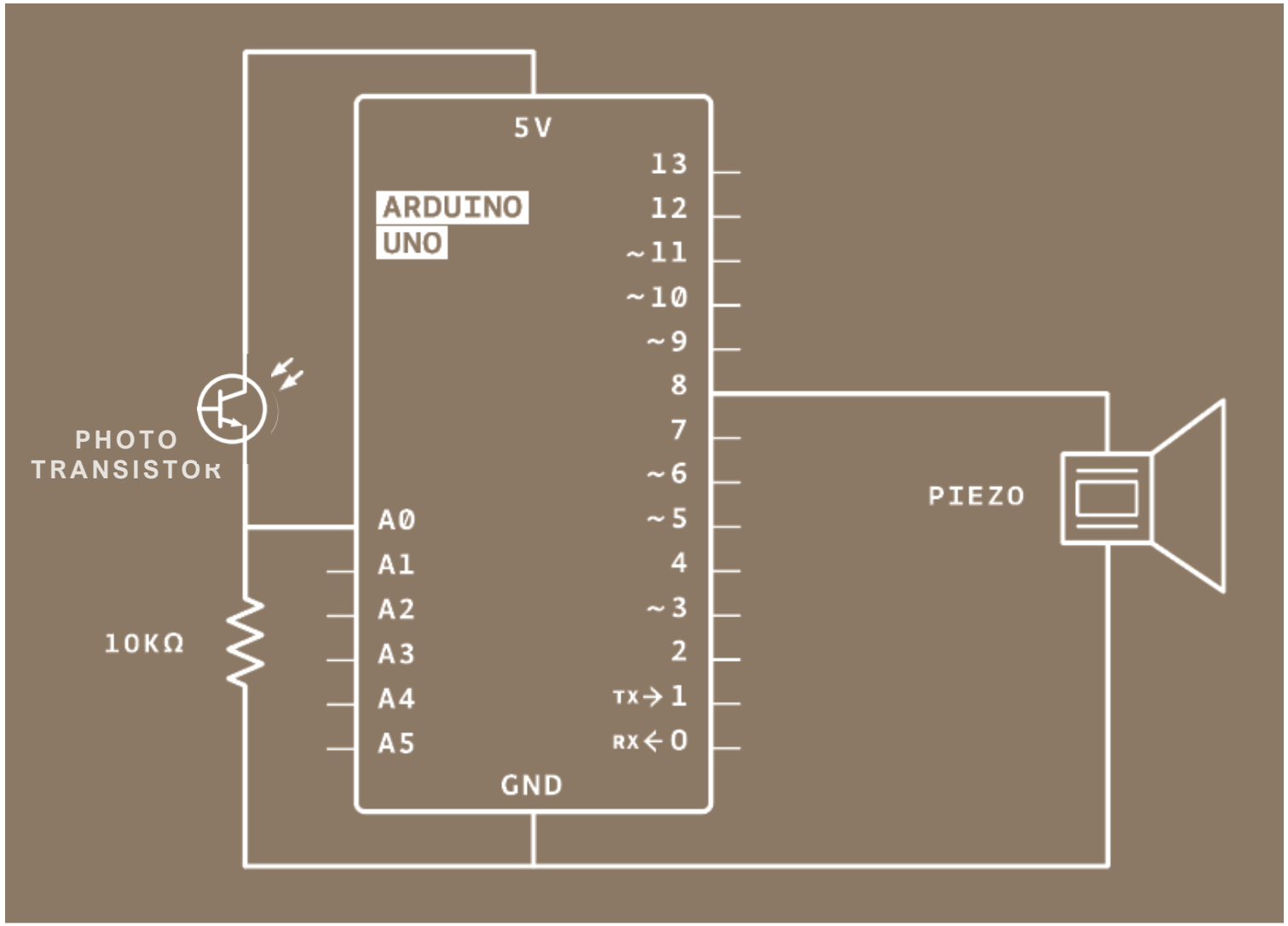
Project Ingredients & Schematic

Time: **45 MINUTES**

Level: ■■■■□

Builds on projects: **1, 2, 3, 4**





Making Sound

tone()

Syntax

```
tone(pin, frequency)
tone(pin, frequency, duration)
```

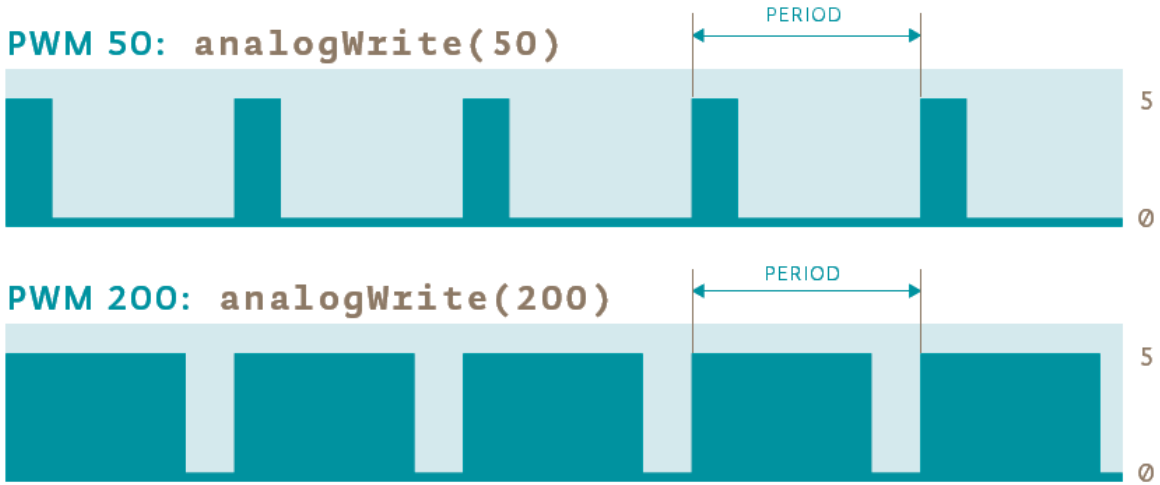
Parameters

- **pin**: the Arduino pin on which to generate the tone.
- **frequency**: the frequency of the tone in hertz. Allowed data types: unsigned int.
- **duration**: the duration of the tone in milliseconds (optional). Allowed data types: unsigned long.

The tone() function makes sound at a specified frequency and duration.

- A transducer, like a piezo, will vibrate at that frequency when it receives the electrical signal, displacing air around it and creating sound.
- The tone() function generates a **square wave** with the specified frequency on a digital pin.

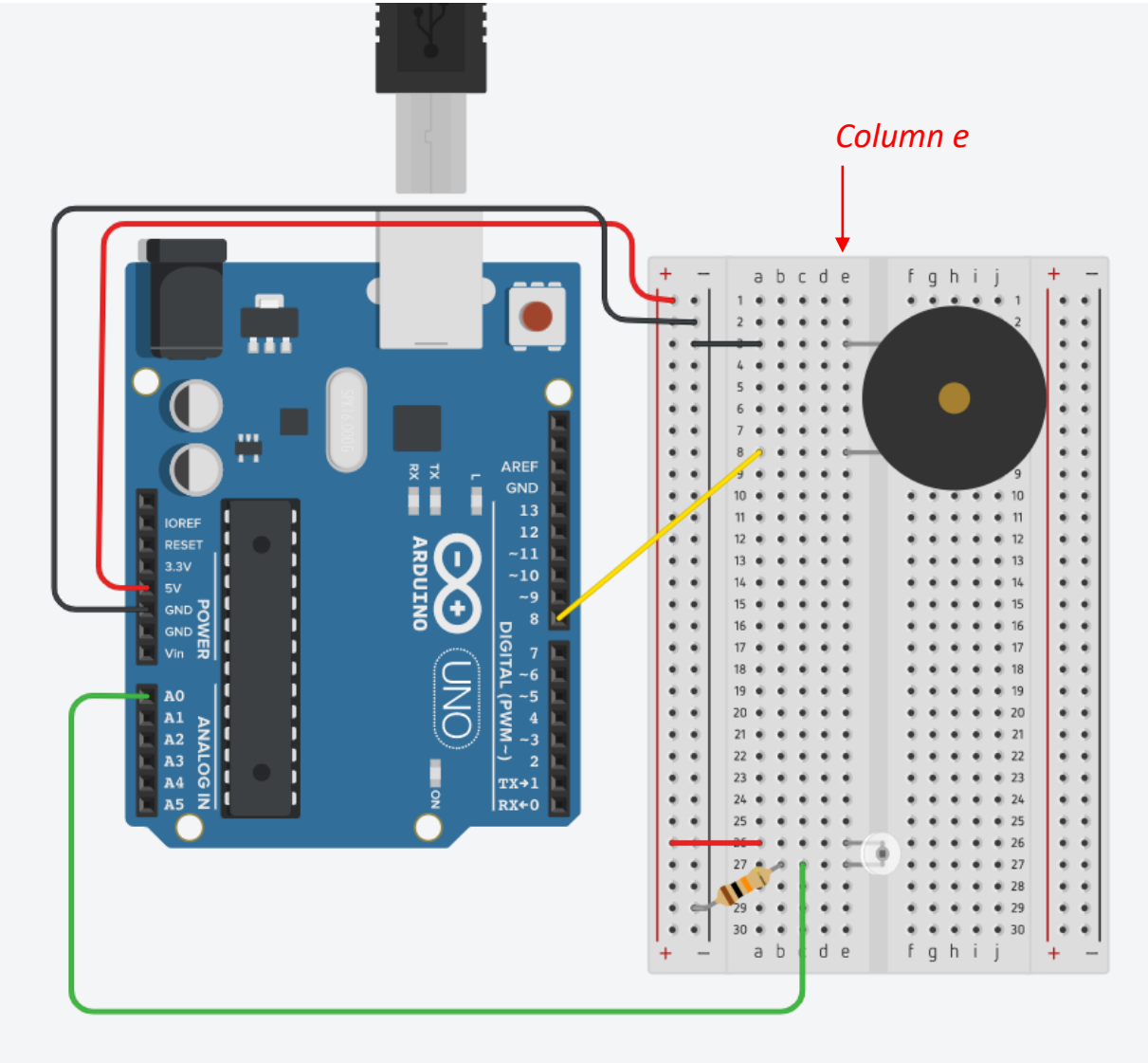
tone() Function



- Note how different PWM values in the **analogWrite()** function maintain the same distance between each wave (period).
- The **tone()** function creates high and low frequency sounds by **changing the period of the wave** while maintaining a 50% duty cycle (the wave is high half of the period.)
- The **tone()** at **880 hz** has the same duty cycle but has a period half as long as the **tone()** at **440 hz**.

$$frequency = \frac{1}{Period}$$

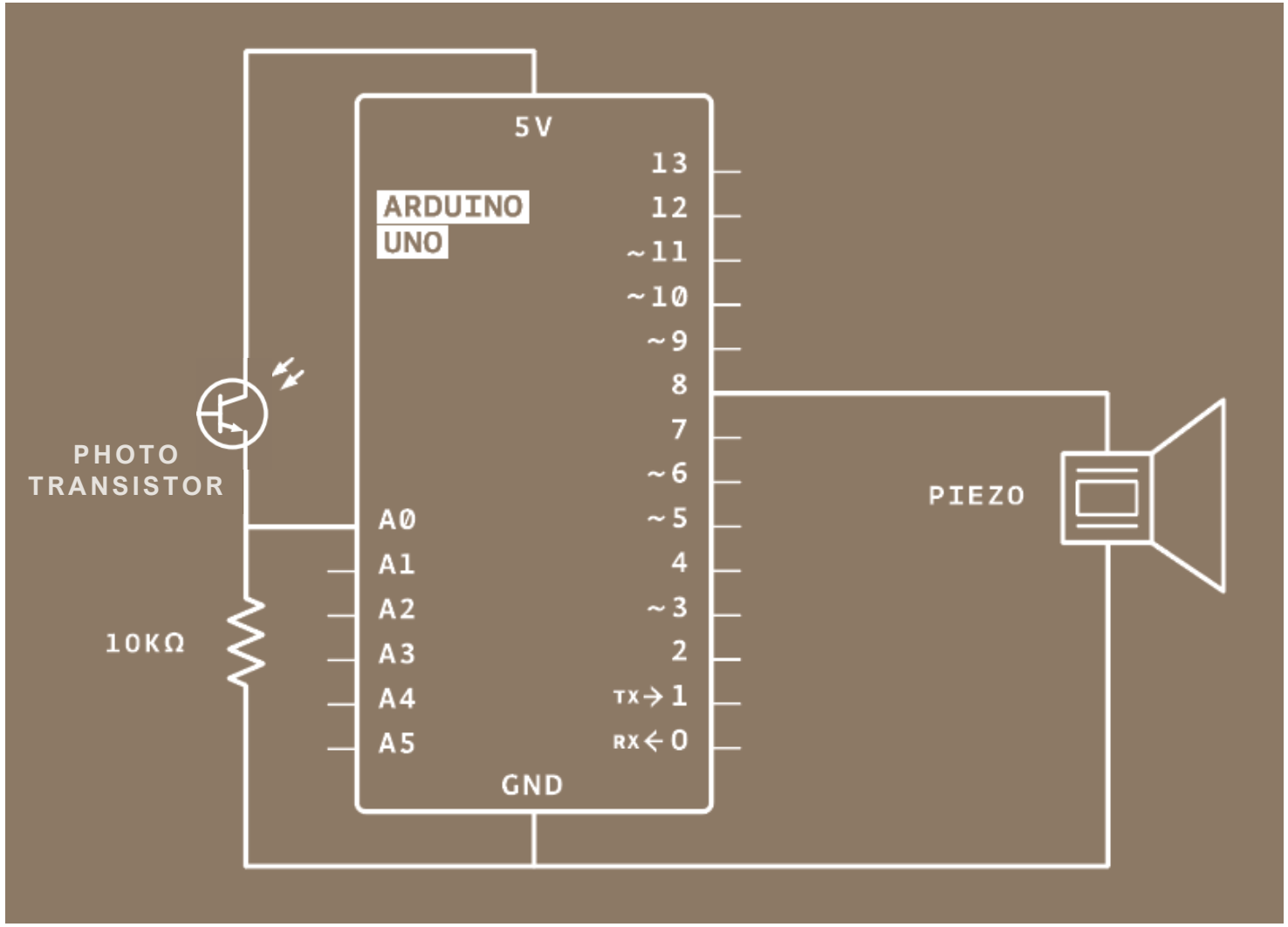
Step 1: Building the Circuit



Building Tips

- Be careful with the legs of the **piezo**. They are easy to bend.
- First place the wires on the breadboard connecting the piezo to the Arduino.
- Then place the **piezo** in *column e* on the breadboard.
- The **phototransistor** is polarized so make sure you have it connected properly.

Compare the Circuit Illustration to the Schematic



This is a simple schematic:

- The **phototransistor** is the analog **input**.
  - Analog pin A0 receives the input voltage values from the phototransistor.
- The **piezo transducer** is connected as a digital **output**.
  - The voltage from analog pin A0 is used to determine the frequency to send to the piezo via digital pin 8.

Step 2: Code Overview

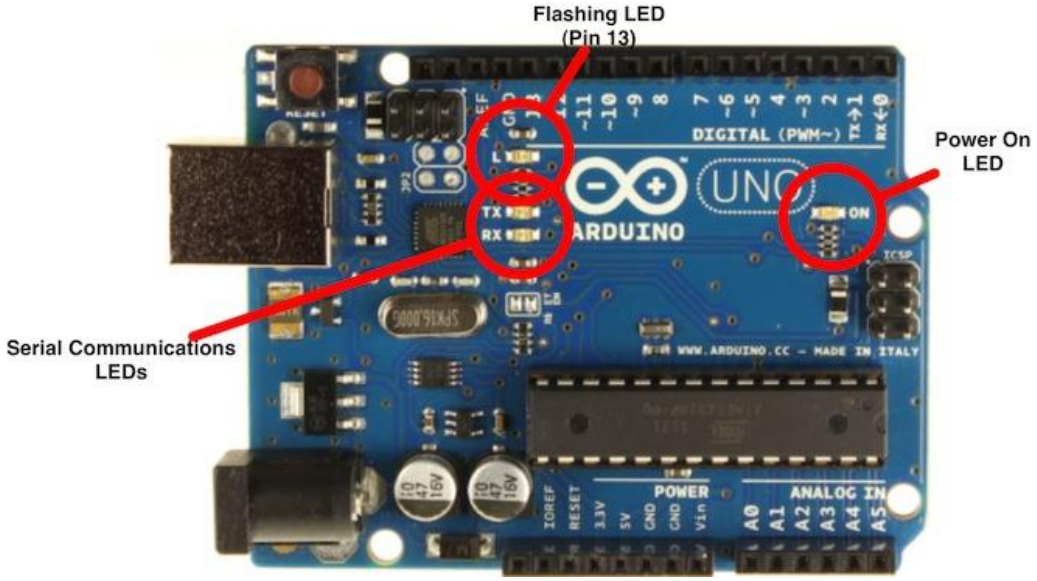
```
int sensorValue;
int sensorLow = 1023;
int sensorHigh = 0;
const int ledPin = 13;

void setup() {
  pinMode(ledPin, OUTPUT);
  digitalWrite(ledPin, HIGH);

  while (millis() < 5000) {
    sensorValue = analogRead(A0);
    if (sensorValue > sensorHigh) {
      sensorHigh = sensorValue;
    }
    if (sensorValue < sensorLow) {
      sensorLow = sensorValue;
    }
  }
  digitalWrite(ledPin, LOW);
}

void loop() {
  sensorValue = analogRead(A0);
  int pitch = map(sensorValue, sensorLow, sensorHigh, 50, 4000);
  tone(8, pitch, 20);
  delay(10);
}
```

Step 2: Code the Project



Syntax

```
while (condition) {
    // statement(s)
}
```

Parameters

condition: a boolean expression that evaluates to TRUE or FALSE.

- The built-in LED light on the Arduino connected to pin 13 is used to indicate when the phototransistor is done calibrating.
- A **while()** loop will repeat continuously until the expression inside the parentheses becomes **FALSE**.
- In this project the **while()** loop will repeat for 5000 milliseconds, or 5 seconds.
- During this time the Arduino is determining the **max** and **min** values from the phototransistor.

Step 2: Code the Project

Syntax

map(*value, fromLow, fromHigh, toLow, toHigh*)

Parameters

value: the number to map.
fromLow: the lower bound of the value's current range.
fromHigh: the upper bound of the value's current range.
toLow: the lower bound of the value's target range.
toHigh: the upper bound of the value's target range.

For example:

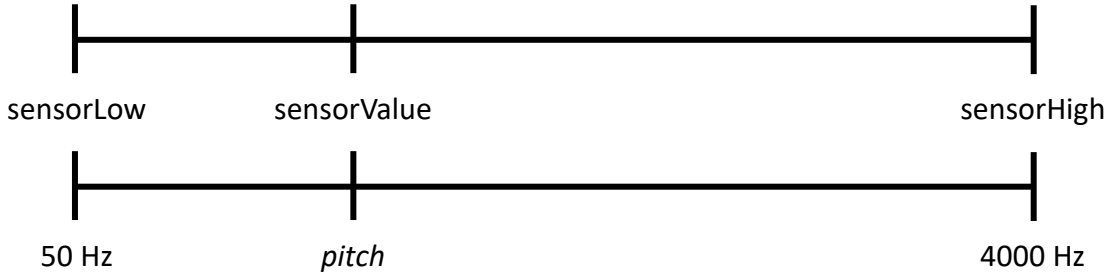
To find a ***pitch*** between 50 Hz and 4000 Hz, if **sensorLow** and **sensorHigh** are calibrated to 5 and 1020, and **sensorValue** was 575, the ***pitch*** would be calculated as follows:

$$\frac{sensorValue}{sensorHigh - sensorLow} = \frac{pitch}{4000 - 50}$$

$$\frac{575}{1020 - 5} = \frac{pitch}{4000 - 50}$$

pitch = 2,237.68 Hz

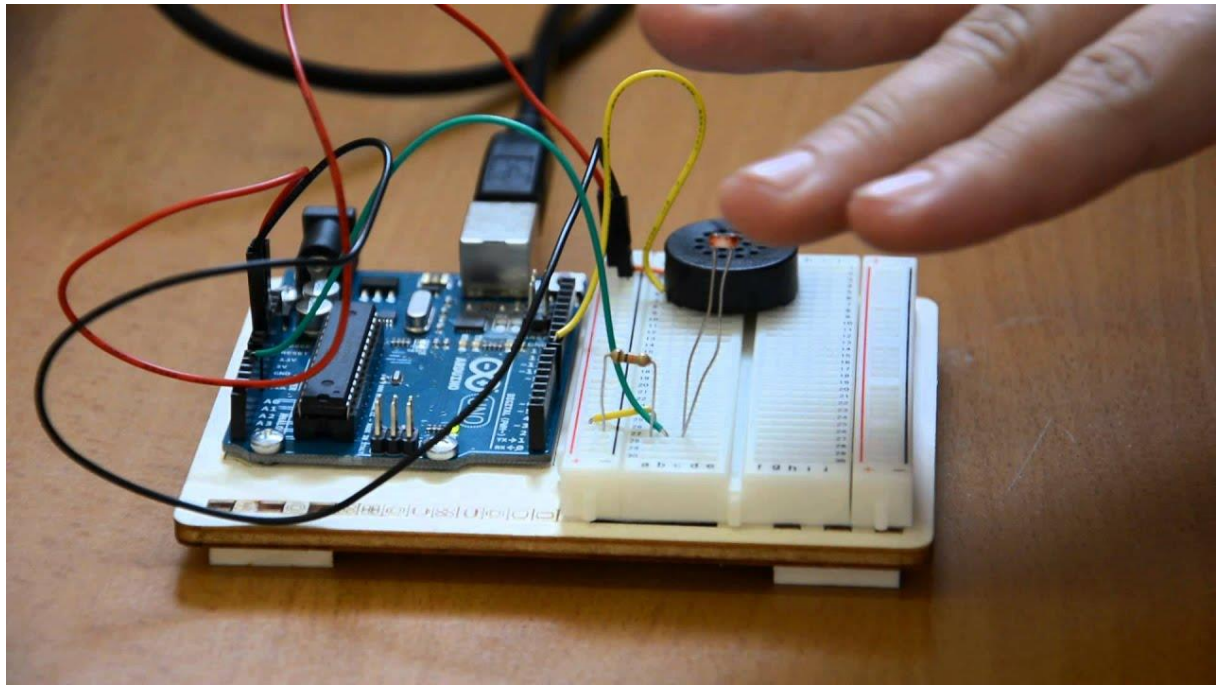
- The **max** and **min** values from the phototransistor are used to determine frequency range for the piezo using the **map()** function.
- The **sensorLow** and **sensorHigh** values will be between 0 and 1023.

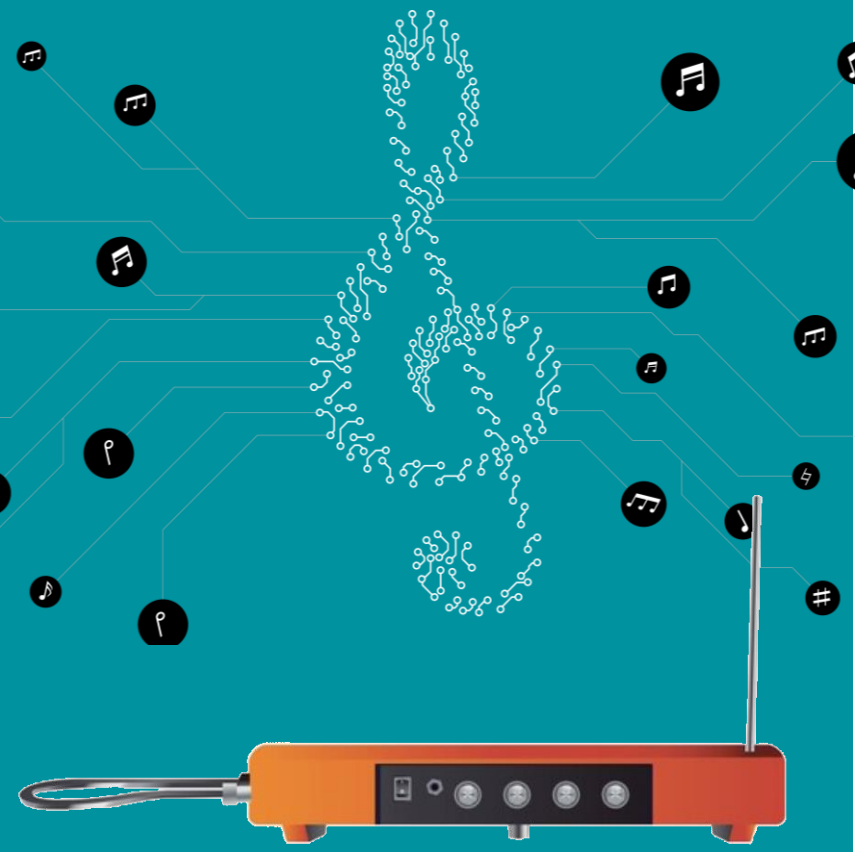


Step 3: Test Your Code

Test your circuit and code once you have completed everything.

- During the 5 second calibration period move your hand back and forth over the phototransistor.
- When the on-board LED turns off and the calibration is over you should hear different frequencies coming from the piezo speaker as you move your hand up and down over the phototransistor.
- **Congratulations! You've made a musical instrument with the Arduino.**





Project 6:

Move over Mozart!
There’s a new Maestro in town!

You completed the Light Theremin!

Questions?